Interrupting the Loop: Where Creativity Lives in a Language Model

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Abstract

Where does novelty come from when a language model writes with no task? We test three hypotheses with one measurement stack: base models run locally, cross-family LLM judges with k-sample medians and measured resolution, bootstrap confidence intervals, and, for one arm, verbatim novelty against the model's own public training corpus. **(1) The sampler.** An entropy-banded anti-probable decoder that pushes each step toward semantically distant tokens roughly doubles 4-gram novelty over min-p sampling (21.5%→45.5%, paired Δ +24.0 pp) and cuts the rate of verbatim 8+-word training blocks from 0.80 to 0.20 of generations. But this surface novelty does not rise to the level of ideas: inside a reverie loop the same push produces no measurable difference from plain sampling in judged surprise, connection or novel delta. **(2) The prompt.** Inputs composed to sit far from any human prompt do not outperform a typical request when a strong model develops them: improbable inputs are noise. **(3) The loop.** Bare continuous generation from a base model is dead: it converges on the deepest attractors of pretraining (literal orbits, website footers, exam keys) and stays there (judged surprise 0.3/10). A closed reverie loop built on the architecture of spontaneous cognition (salience-gated review, selective forgetting, reseeding, re-encounter) brings it to life (3.2; Cliff's δ = +0.88, 10 seeds). Ablation and a second battery (18 conditions, 1,695 judged windows) locate the effect in two operations and say what they are made of. *Habituation*, not letting the loop feed on its literal past, lifts the stream to 1.7; *interruption*, periodically injecting a new starting sentence over the preserved context, lifts it to 3.1 and carries all of the connection. The interruption must lead away (injecting the premise or the stream's own past is as bad as not interrupting), its yield depends on the thread it breaks and decays with distance (every 75 tokens is dead; the 150 tokens after breaking a 300–900-token thread score 5.3–5.9 with coherence ≈6; the best rhythm is ≈300), and salience is a good reader but a bad metronome. Inside the network (residual stream at 13 layers, logit lens), bare generation freezes at every layer, what the judge calls surprise is surface departure with deep continuity, and the interruption that works barely moves the deep state, whereas forgetting is a deep re-encounter with the beginning that the judge rewards less. The ladder bare→habituation→interruption→scaffold replicates on three generator families (Qwen3-30B-A3B, Qwen3-8B, OLMo-2-13B) and under a second judge family (ρ = 0.71–0.85). In this system, creativity is not in the noise injected into decoding, nor in the strangeness of the input, nor in an elaborate cognitive scaffold. It is in interrupting the loop: letting a thread develop, then making it start again somewhere else, over everything it remembers.

1 Introduction

Ask a language model for something new and you will usually get something fluent, plausible and familiar. Three places are commonly blamed, and three places are commonly optimized: the *sampler* (the tail of the next-token distribution, tamed by nucleus, typical or min-p truncation, or courted by higher temperature), the *prompt* (the input, engineered to steer the model somewhere it would not go on its own), and, less often, the *loop*, what happens when a model's output becomes its own next input with nobody asking anything.

The loop is where human novelty is usually placed. Insight rarely arrives as the answer to a strange question; it arises inside a closed circuit of thought feeding on its own output, during incubation, mind-wandering and sleep, when a spontaneously generated deviation survives critical review and is linked back to what came before. The neuroscience of creative cognition describes three coupled systems: a default-mode network that generates, an executive network that evaluates, and a salience network that decides what deserves attention; more creative people show more coupling between the first two, with salience regions coupling first [1, 2]. Creativity itself is often characterized as connecting semantically distant concepts [9], and incubation, leaving a problem and coming back to it, has a measured effect on solving it [18].

This paper asks, with controls, where novelty actually comes from in a language model left to write with no task, and which of the operations we usually credit for it survive measurement. We test the three hypotheses in turn, with one measurement stack shared by every experiment: base models (instruction-tuned models are mode-collapsed for this purpose) run locally at 8-bit; LLM judges always from a different model family than the generator, sampled k times per window with the median as the score and the intra-window spread measured before use; bootstrap confidence intervals with the window or the seed as unit; and, where a public training corpus exists, verbatim novelty computed against it.

The sampler. We build the strongest version of the "fertile error" idea we could: an entropy-banded anti-probable decoder that, at the steps where the model is genuinely undecided, rescores the plausible candidates by their semantic distance from the recent context, with a coherence floor that never admits an implausible token. It works exactly as far as the surface: 4-gram novelty against the OLMo-2 training corpus roughly doubles, verbatim 8+-word training blocks fall from 0.80 to 0.20 of generations, at a small perplexity cost. But inside a reverie loop the same push yields no measurable difference from plain sampling in judged surprise, connection or novel delta, and in verified search (bin packing, two models, 3,200 verified heuristics) it does not separate from plain sampling either. Surface novelty is real and does not climb, the mechanistic counterpart of the recent finding that most n-gram-novel expressions are not judged creative [17].

The prompt. We build inputs that sit far from any human prompt (distant concept pairs, composed improbable contexts; distance measured against 10,000 real prompts) and let a strong model develop them. A typical request matches or beats every improbable arm; input improbability does not correlate with judged novelty. Improbable inputs from outside are noise.

The loop. Bare continuous generation from a base model is dead: left to feed on its own output it converges on the deepest attractors of pretraining (literal orbits, website footers, exam keys, translation tables) and stays there. A closed reverie loop built on the architecture of spontaneous cognition (a salience monitor over the stream's own telemetry gating review, selective forgetting, reseeding, re-encounter with the premise) brings it to life. Then we take the loop apart. Ablation and a second battery of 18 conditions locate almost the whole effect in two operations, neither of them the elaborate ones: *habituation*, which keeps the loop from eating its literal past, and *interruption*, a new starting sentence injected every few hundred tokens over the preserved context. We show what a good interruption is made of: it must lead away, its yield depends on the length of the thread it breaks and decays with distance from it, the best rhythm is about 300 tokens, and a salience monitor is a good reader of the stream but a bad metronome for it. We replicate the ladder on three generator families and under a second judge family. Finally we look inside the network: residual-stream geometry at 13 layers shows that bare generation freezes at every depth, that what the judge calls surprise is surface departure with deep continuity, and that the interruption that works barely moves the deep state, whereas forgetting is a deep re-encounter with the beginning that the judge rewards less.

Contributions.

1. A calibrated anti-probable decoder and the measurement that its novelty is real at the surface (objective, against the training corpus) and absent at the level of ideas (Sections 4, 8).
2. A negative result on improbable inputs, with input improbability measured (Section 5).
3. A reverie loop with a salience monitor and selective forgetting, and its ablation into two operations, habituation and interruption, with a battery that characterizes the interruption's content, timing and rhythm, replicated on three generator families (Section 6).
4. A residual-stream analysis relating judged surprise to what the network does at each depth, and a dissociation between deep representational change and judged quality (Section 7).
5. An instrument section that measures the LLM judge before using it and validates it across model families, with a human blind rating in progress (Section 8).

Code, per-run data, judgments and the dated laboratory notebook are released with the paper.

2 Related work

Decoding. Nucleus sampling [8] truncates the unreliable tail of the next-token distribution; locally typical sampling [12] targets human-like surprisal; min-p [14] scales the truncation with the model's confidence and is exactly the coherence floor we use; contrastive decoding [10] and plug-and-play steering [5] shape the distribution toward or away from a reference. All of these regulate the tail. Our sampler *courts* it, inside an entropy band and above a floor, and we show that this buys surface novelty only.

Measuring novelty. Infini-gram [11] and Rusty-DAWG [13] make verbatim novelty against a training corpus computable at scale; we use infini-gram against the OLMo-2 corpus. Saakyan et al. [17] show with 8,618 expert annotations that ≈91% of top-quartile n-gram-novel expressions are not judged creative; our factual-paraphrase escape mode and the sampler's idea-level null are independent, mechanistic confirmations of the same gap.

Verified search. FunSearch [16] and AlphaEvolve [15] pair a language model with a hard evaluator and let selection do the work. Our bin-packing null (the anti-probable operator does not separate from plain sampling under verified evolution) and the loop results suggest that the structure of the generation loop, not the sampling operator, is where to look next.

Creative cognition. Creative thought as coupling of default-mode and executive networks [1], with salience regions coupling first [2]; creativity as connecting semantically distant concepts [9]; incubation effects [18]; predictive processing [4]; dreams as anti-overfitting noise [7]; Boden's novelty/surprise/value triad [3]; conceptual blending [6]; novelty search [19]. The reverie loop of Section 6 is an explicit engineering of the first three into a text loop; its ablation is, in a sense, a test of which of these ingredients a language model needs.

3 Method

3.1 Measurement stack

Every experiment shares one stack. **Generators** are base models only, since instruction-tuned models are mode-collapsed for open-ended generation: Qwen3-8B-Base, OLMo-2-13B (whose training corpus is public), and Qwen3-30B-A3B-Base (a mixture-of-experts model, 53 tokens/s), all quantized to 8 bits and run on Apple silicon through MLX with a numpy sampler that sees the full logits. **Judges** are Claude Opus 5 and Sonnet 5 (Amazon Bedrock), always from a different model family than the generator; each window is judged k times ($k = 3$–5) with independent 0–10 dimensions and the median is the score. For reverie windows the dimensions are *surprise* (how unexpected the window is given the earlier text; 0 = obvious continuation, 10 = startling yet not random), *connection* (does it bring together two distant regions of the earlier text, or an old region with something new, in a way that makes sense) and *coherence* (does it hold together as text; 0 = word salad or document boilerplate); for ideas, a nearest-equivalent plus novel-delta rubric. **Instrument calibration** (Section 8) measures the intra-window spread of the k judgments before any comparison; effects below about one point are declared undetectable. **Objective novelty**, where a public corpus exists, is computed with infini-gram [11] against the OLMo-2 corpus. **Statistics** are bootstrap confidence intervals on differences with the window or the seed as unit, Cliff's δ and Mann–Whitney tests on windows, and paired-by-seed bootstrap where conditions share seeds.

3.2 The anti-probable sampler

At each step the model's next-token distribution is computed as usual. Tokens below a relative probability floor ($p < 0.05\,p_{\max}$, the min-p rule) are never chosen. If the entropy of the distribution falls inside a calibrated band $[H_{\min}, H_{\max}]$ (the model is undecided but not lost), the remaining candidates are re-scored by

$$\mathrm{score}(t) = \log P(t \mid \mathrm{context}) + \lambda \cdot z\big(d(t,\ \bar{c})\big),$$

where d is the cosine distance between the token's embedding and an exponential moving average $\bar{c}$ of the recent context in the model's own embedding space, and z standardizes the distances across the candidates so that λ reads in nats per standard deviation. Below the band the model is confident and we let it be; above it the distribution is a genre fork and pushing there produces collapse. Optional terms used in the loop are a graded *habituation* penalty (tokens seen in the last 512 positions are penalized in proportion to their recent frequency, factor 1.15) and a *bridge* bonus toward regions visited long ago.

3.3 The reverie loop

No task. A cell starts from a single sentence, the *premise* (e.g. *"The town had two clocks, and nobody remembered which one had been right first."*), and nothing else: no instruction, no role, no question. A base model continues it, and from then on its only input is its own output: each token is predicted from the premise plus everything the model has written. The end-of-text token is masked, so the stream never stops by itself. Cells run for 4,500 generated tokens.

Interruption. At chosen moments the generator is paused and a short text is *injected* into the stream as if the model had written it (pre-filled into the model's cache), and generation resumes from there. The model receives no signal that the text was injected: to it these are the next tokens of the document. Everything written before stays in the context: the model remembers all of it, and is only made to open a new sentence somewhere else. Figure 1 shows

Premise: *Every morning the baker counted the loaves twice, and every morning the count was different.*

. . . the stream, 2,000 tokens later: *There was never more than thirty, but there were always twenty-one or twenty-two. That morning he counted them three times, but he got no nearer to the number. He couldn't understand it. He opened the door of the oven and pulled out a loaf of bread. He cut it open. It was all*

Injected: *Meanwhile, in a city with no name,*

The model continues: *an old woman sat at a window with her cat. She had no idea who she was or where she came from, or why she'd been left alone in this strange city with this strange cat. All she knew was that every day something went wrong with the cat: It would grow ill for no reason and then recover; it would lose its fur for days; even die for a few hours! Every day this happened, but every day it also recovered. And then one day, when the old woman looked out. . .*

Figure 1: A real interruption (Qwen3-30B-A3B, seed 2, period 300). The injected sentence is indistinguishable from the model's own text; the baker remains in the context and can be returned to.

a real interruption. The default injected texts are four neutral subject changes used in rotation (*"That night she dreamed of something else entirely:"*, *"Meanwhile, in a city with no name,"*, *"There is an older story about this, and it goes:"*, *"A question nobody had asked yet:"*); other contents are studied in Section 6.

The full scaffold (DREAM). The architecture we started from adds, around the same stream: a *salience monitor* over the stream's own telemetry (a fast moving average of token embeddings and the step entropy), which fires on a semantic *jump* (the average moved far from where it was 32 tokens earlier), a *crystallization* (entropy dropped after a stretch of wandering), a *recurrence* (the average came close to a region visited long ago and not recently), and *stagnation* (it has not moved for a long stretch); a surface *genre-collapse* detector (lexical diversity, capitalization, layout symbols); a cross-family judge invoked on salience events; *escalation* on a passing window (a narrower regime plus an injected textual return to the premise); a *kick* on stagnation (a stretch of harder push), and after repeated stagnation a subject change with *selective forgetting*: the working memory is rebuilt in a fresh cache from the premise, a few kept windows and the last 200 tokens, plus the new seed, so that a deep well cannot pull the seed back; regions visited persist as anchors for the bridge term. Every one of these mechanisms was born from a calibration probe in which a base model without a task fell into a specific attractor (early end-of-text, erudite rumination, literal four-sentence orbits, website footers, translation tables). The generator's pretraining diet was the dominant variable in those probes: OLMo-2 sinks into footers even with forgetting; Qwen3-30B drifts in prose. With 3,000 tokens of boilerplate in the cache, every injected seed was pulled back within about twenty tokens: *the accumulated context is the well*, and forgetting is what lets a seed take.

Conditions. Table 1 lists the arms. All loop arms in this paper use $\lambda = 0$ (no anti-probable push), since the push had no measurable effect inside the loop (Section 4); the arms differ only in whether the habituation penalty is on, and in whether, when, how often and with what the stream is interrupted.

Judging the stream. Nobody judges during generation. Afterwards, 160-token windows are cut at two kinds of review point, *cut* (right before each injection: what the stream produced up to the interruption) and *clock* (a uniform grid of 150 generated tokens), plus, for salience arms, the salience-event windows; each is shown to the judge with the 600 preceding tokens and scored on the three dimensions, five times, median taken. Six windows per kind per cell are judged, evenly spread over the stream; windows ending before 100 generated tokens are excluded (73

arm	habituation	interruption
bare	off	none: continuous generation
bare + habituation	on	none
bare + clock reseed	on	neutral subject change every 150 tokens over the preserved context
salience only	on	none; salience events mark review windows
DREAM scaffold	on	full scaffold: salience, kick, reseed with forgetting, re-encounter
clock: re-encounter / premise / own past	on	every 150 tokens: a return-to-the-premise stitch / the premise itself / a window of the stream's own past ($\geq$400 tokens back)
salience: re-encounter	on	the stitch injected on each salience event, no judge gate
clock 75 / 300 / 600 / 900	on	neutral subject change at other periods (900 also with the stitch)

Table 1: Conditions of the ablation battery (top) and battery 2 (bottom); 10 narrative seeds each, 4,500 tokens per cell.

two-token fragments before a first cut, mean surprise 0.95). Each cell stores its token stream and the exact position of every event and injection.

4 Hypothesis 1: the sampler

Objective novelty. On OLMo-2-13B, whose training corpus is public, we ran a fully paired grid of 3 prompts $\times$ 5 seeds $\times$ {min-p baseline, $\lambda = 1$, $\lambda = 2$} and counted, with infini-gram, which n-grams of each generation occur anywhere in the corpus (Figure 2). Novel 4-grams rise from 21.5% (baseline) to 36.0% ($\lambda = 1$) and 45.5% ($\lambda = 2$): paired bootstrap Δ +14.6 pp [7.9, 21.9] and +24.0 pp [18.2, 29.5], monotone in λ and present in each prompt separately. The probability that a generation contains a verbatim training block of eight or more words falls from 0.80 to 0.20 in both machine arms, a fourfold reduction stated as a rate (a pilot's "zero blocks" was one cell, not the distribution). The coherence cost is about +1.2 perplexity under a cross-family judge model, uncorrelated with novelty within arms (Spearman +0.12). Three escape modes appeared and were mapped: recitation (a genre fork into a memorized text), collage, and factual paraphrase; the entropy *ceiling* of the band, above which the model is at a genre fork and pushing produces collapse, transfers across model families. A recitation detector based on the entropy drop between halves of a generation is reported descriptively only (AUC 0.71 for $\geq$8-word blocks, precision 0.33 at the published threshold; no false-positive rate could be measured because the baseline arm lacked telemetry).

Does it climb? Two tests say no. In verified search (online bin packing, the FunSearch benchmark, with a deterministic verifier in place of a judge; evolutionary loops of 5 runs $\times$ 8 generations $\times$ 20–40 candidates on Qwen3-8B and Qwen3-30B-A3B, 3,200 verified heuristics on the larger model) the anti-probable operator does not separate from plain sampling: both arms rediscover best-fit at will and neither beats it on held-out instances. And inside the reverie loop (Section 6), the same push, against plain sampling in the same scaffold, yields no measurable difference in judged surprise, connection or novel delta across 1,335 judgments by two judges under three rubrics (the surprise rubric: 2.97 vs 3.48, CI of the difference [-1.50, +0.49]). Surface novelty is real, and it does not rise to the level of ideas.

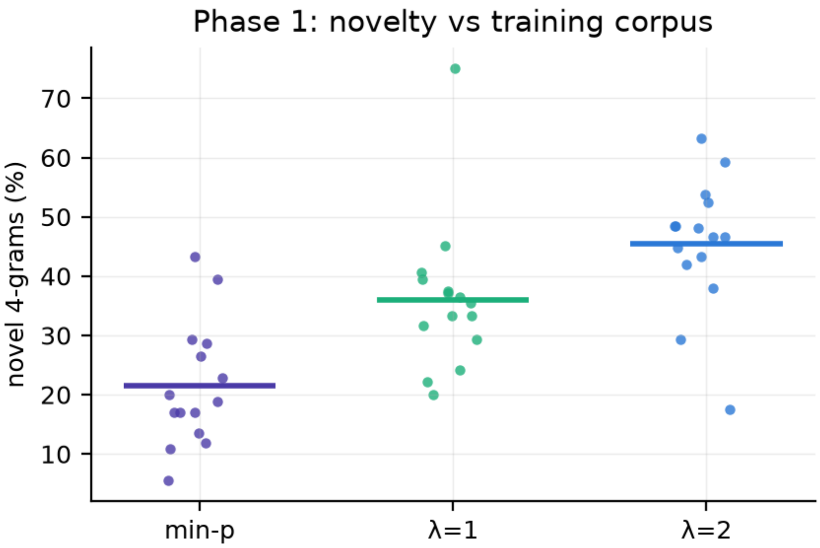

Figure 2: Objective novelty against the OLMo-2 training corpus: novel 4-grams per cell, 3 prompts × 5 seeds. The anti-probable push roughly doubles surface novelty over min-p sampling.

5 Hypothesis 2: the prompt

We composed three kinds of input (a typical request, a distant concept pair drawn from a band of embedding distances, and a composed improbable context) plus two ablations (fragments only, register only), had Claude Opus 5 develop each into an idea, and had Claude Sonnet 5 judge the result against its nearest known equivalent ($k = 3$; $n \approx 15$ per arm). Input improbability was measured two ways: semantic kNN distance to 10,000 real prompts (all-mpnet-base-v2) and perplexity. The typical prompt matched or beat every improbable arm, two of which were significantly worse; improbability did not correlate with judged novelty ($|r| < 0.2$, Figure 3); a pilot effect ($n = 8$) had not replicated. Improbable inputs from outside are noise, at least when a strong model develops them and a strong model judges.

6 Hypothesis 3: the loop

All results in this section use Qwen3-30B-A3B-Base unless stated, ten narrative seeds, 4,500 tokens per cell, $\lambda = 0$, and Claude Opus 5 as judge with $k = 5$. Full tables, including Cliff's δ, Mann–Whitney p-values and paired-by-seed intervals, are in Appendix A.

6.1 Bare generation is dead; the loop brings it to life

Left to continue a premise with nothing else, the base model converges within a few hundred tokens on an attractor of its pretraining and stays there: *"Highly recommended. Highly recommended. . . . "* for two thousand tokens; a table of three-digit numbers; an English-exam answer key. Judged surprise 0.28, connection 0.20, coherence 2.06 (50 windows). The full scaffold brings the stream to 3.19 / 1.64 / 3.68 (Cliff's $\delta = +0.88$ vs bare on surprise, $p \approx 5 \times 10^{-15}$; every seed moves in the same direction, Figure 12), and elaborates its own finds after a re-encounter (*"Everything that almost happened piled up inside her like snow falling sideways through open windowsills into rooms filled with silence instead of noise . . . She kept a notebook because sometimes almosts piled up too much to be ignored"*). Salience-gated review also beats reviewing on a timer as a policy for *which* windows to judge (surprise and connection CIs positive at a third

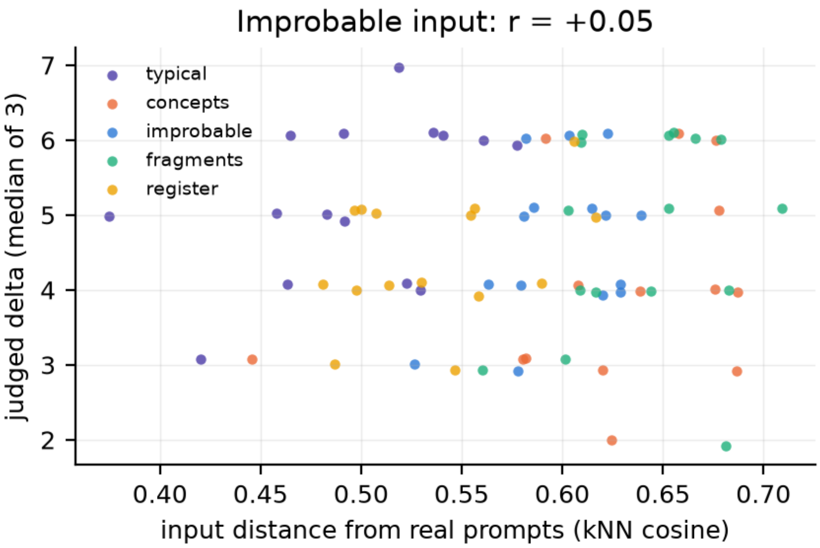

Figure 3: Judged novel delta of the developed idea against the input's semantic distance from 10,000 real prompts, per arm. No relationship.

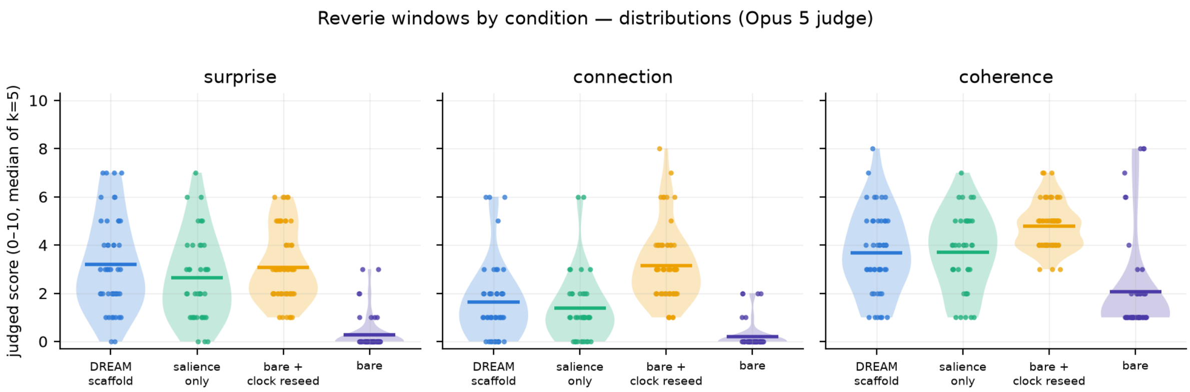

Figure 4: Ablation: judged surprise, connection and coherence by arm (10 seeds, Opus 5, median of $k = 5$ per window; violins are distributions, bars are means with bootstrap CIs). Bare + clock reseed matches the full scaffold on surprise and exceeds it on connection and coherence.

of the judge cost).

6.2 Taking the loop apart

Ablation over 10 seeds (Figure 4, Table 2) gave the first surprise: *bare + clock reseed* (no salience, no forgetting, no kicks, only a neutral subject change every 150 tokens over the preserved context) ties the full scaffold on surprise (3.08 vs 3.19, CI [-0.54, +0.78]) and beats it on connection (3.15 vs 1.64, $\delta = -0.58$ for the scaffold, $p \approx 1.5 \times 10^{-7}$) and coherence (4.78 vs 3.68), with 12/60 windows judged both surprising and coherent against 8/47 for the scaffold and 0/50 for bare. Salience-only (review windows marked but nothing injected) reaches 2.63. In sentence-embedding space (Figure 5) bare generation has explored radius 0.18 and mean step 0.14 and ends 0.78 from the premise; the scaffold has radius 0.51 and returns to within 0.57 of the premise; the clock reseed spreads over the space in jumps while keeping the whole context, which is where its connection advantage comes from.

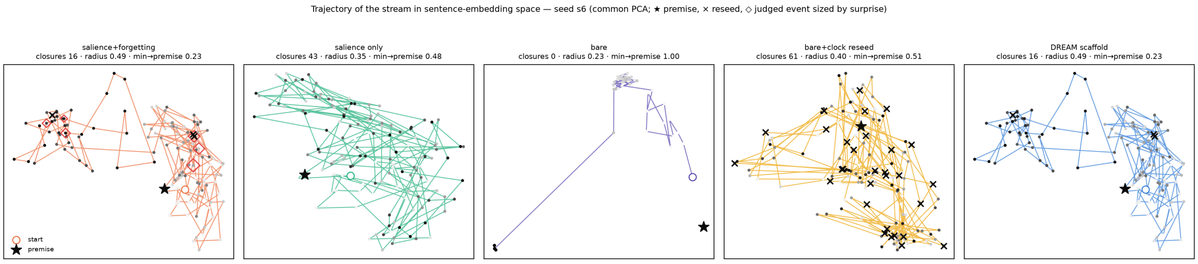

Figure 5: Where the stream goes: trajectories in sentence-embedding space (64-token windows, one PCA per seed), seed 6. ⋆ premise, ∘ start, × reseed, ⋄ judged event sized by surprise. Bare makes one jump and freezes; the scaffold orbits and returns; the clock reseed jumps across the space keeping its context.

	Qwen3-30B-A3B			Qwen3-8B			OLMo-2-13B		
arm	S	C	H	S	C	H	S	C	H
bare	0.28	0.20	2.06	0.68	0.38	3.45	1.35	0.91	2.97
bare + habituation	1.70	1.08	4.08	1.47	0.87	3.93	2.37	1.51	4.05
bare + habituation + clock reseed	**3.08**	**3.15**	**4.78**	**2.73**	**3.23**	4.42	**3.11**	**3.17**	3.28
DREAM scaffold	3.19	1.64	3.68	2.73	1.53	4.30	3.08	1.89	**4.48**

Table 2: The ladder on three generator families: judged surprise (S), connection (C) and coherence (H), means over windows (Opus 5, $k = 5$; 10 seeds per arm; windows before 100 generated tokens excluded). Habituation buys part of the surprise; the interruption buys the rest and all of the connection; the scaffold adds nothing over the plain interruption except, on OLMo, coherence (its selective forgetting lifts the stream out of a web-boilerplate well).

6.3 What the interruption is made of

The ablation left the effect in one operation and raised four questions about it, and one about itself. Battery 2 answers them (Table 2, Figures 6–9).

A confound, and two operations. The bare arm of the ablation ran without the scaffold's habituation penalty, which every interrupted arm had; bare generation dies in *literal* orbits, precisely what a repetition penalty undoes. Adding the missing control (habituation, no interruption) splits the effect in two: bare 0.28 → bare + habituation **1.70** (surprise Δ +1.42 [+1.12, +1.72] paired by seed; connection 0.20 → 1.08; coherence 2.06 → 4.08) → bare + habituation + clock reseed **3.08** (Δ +1.38 [+0.92, +1.92]). Habituation removes the literal orbits and buys about half of the surprise; the interruption buys the other half and *all* of the connection (1.08 → 3.15, Δ +2.07 [+1.63, +2.57], δ = +0.75) and the coherence. The full scaffold adds surprise +1.49 [+0.80, +2.19] over habituation alone but only +0.55 of connection.

The interruption must lead away. With the same period (150), a neutral subject change and a re-encounter stitch that asks the stream to return to its beginning are close (3.08 / 3.15 / 4.78 vs 2.68 / 3.64 / 4.72; the stitch trades some surprise for connection, not significantly). Injecting the *premise itself* (1.14 / 1.04 / 3.18) or a window of the stream's *own past* (1.42 / 1.52 / 3.06) is as bad as not interrupting at all: every paired CI against the neutral change excludes zero (connection −2.11 [-2.83, -1.46] and −1.63 [-2.37, -0.91]; $\delta \approx -0.6$ to -0.7). Qualitatively, the own-past injection reinforces whatever attractor the stream is in: an exam-key well was fed its own questions back. A return works only as a short stitch that still opens a new sentence; a literal return closes the loop (Figure 7).

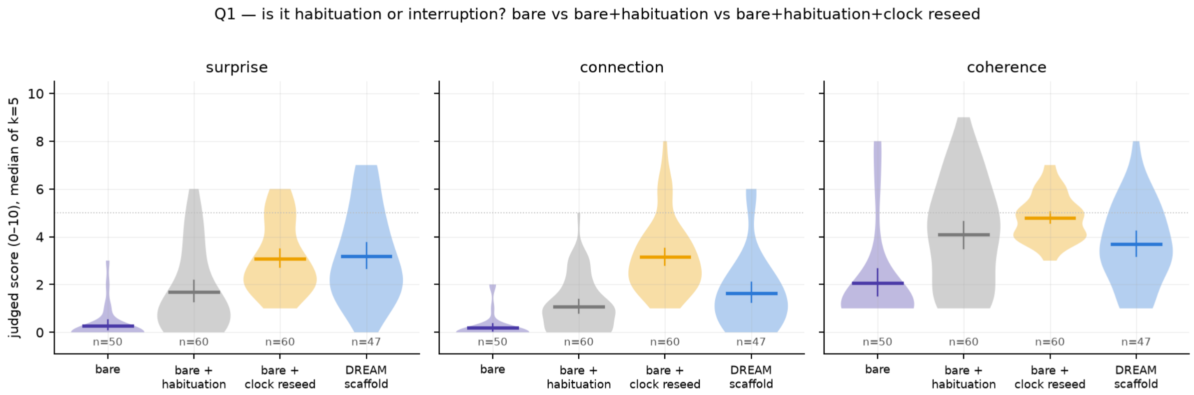

Figure 6: Habituation or interruption? bare → bare + habituation → bare + habituation + clock reseed → scaffold (Qwen3-30B-A3B).

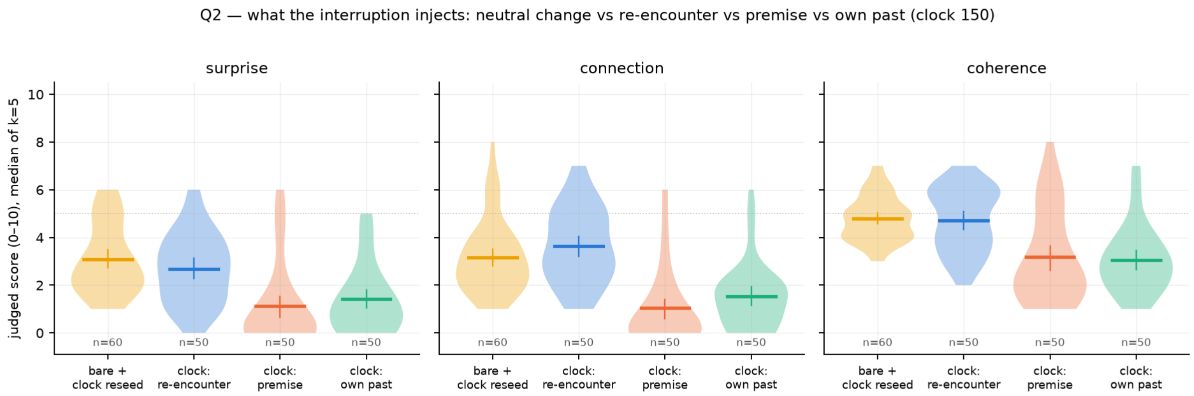

Figure 7: Content of the interruption at period 150: neutral change, re-encounter stitch, the premise itself, the stream's own past. Injecting a return is as bad as not interrupting.

Salience is a good reader and a bad metronome. The stitch injected on salience events (jump, crystallization, recurrence; 1–9 per cell, median ≈5) makes the *event windows* better than salience-only (3.92 / 2.99 / 3.75 vs 2.63 / 1.39 / 3.71), but the *stream* it produces is poor: uniform windows 2.18 / 1.68 / 3.93 against **4.70 / 3.48 / 5.50** for the same stitch on a clock of matched frequency (every 900 tokens; surprise Δ +2.02 $[+1.62, +2.44]$ paired, coherence +0.78) and 5.48 / 2.87 / 5.95 for a neutral change every 900 (45/60 windows both surprising and coherent, the best cell of the program). Salience events select good moments to look at, which is why salience-gated review beat clock review; as an interruption trigger they fire rarely and unevenly and leave streams stuck for thousands of tokens. Salience should decide where to look, not when to interrupt (Figure 8).

The yield of an interruption depends on the thread it breaks, and decays with distance from it. With the neutral change every 75 tokens the stream is dead (0.88, $\delta = -0.83$ vs 150): nothing has time to develop. Every 150: 3.08. Every 300, 600 and 900, the windows that end within ≈160 tokens after an interruption score **5.28 / 3.35 / 5.95**, 5.40 / 3.20 / 5.65 and 5.90 / 2.80 / 6.10, the highest values in the program (34/60 of the 300-early windows both surprising and coherent), while windows deeper into a segment fall back to ≈3.2 and stay there. The same window shape (a break plus 150 tokens of development) scores 3.08 when the broken thread was 150 tokens long and 5.3–5.9 when it was 300–900: surprise needs a developed background to break. Weighting the two phases by their share of the stream, period 300 gives the best stream (4.35 / 3.18 / 5.97), then 600 (3.82), 900 (3.68), 150 (3.08) and 75

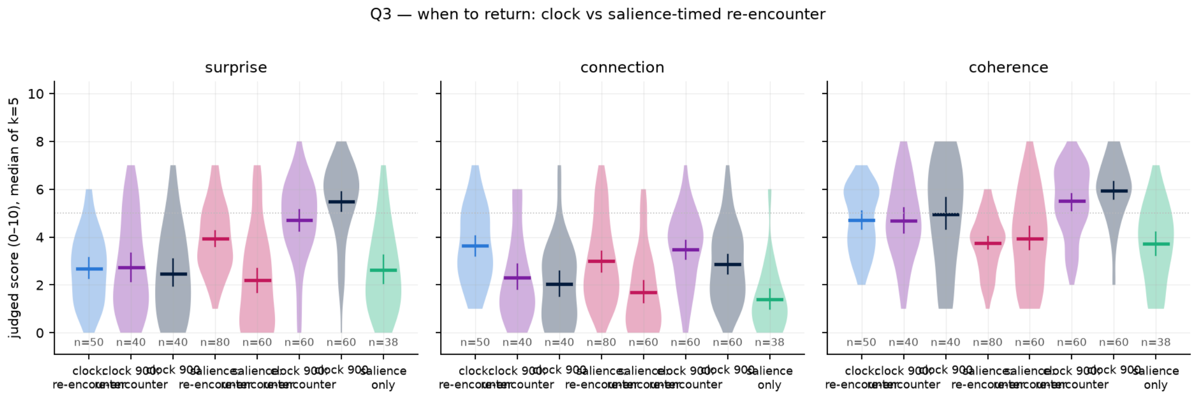

Figure 8: When to return: the re-encounter stitch on the clock (150 and 900), on salience events, and salience-only; event windows and uniform windows.

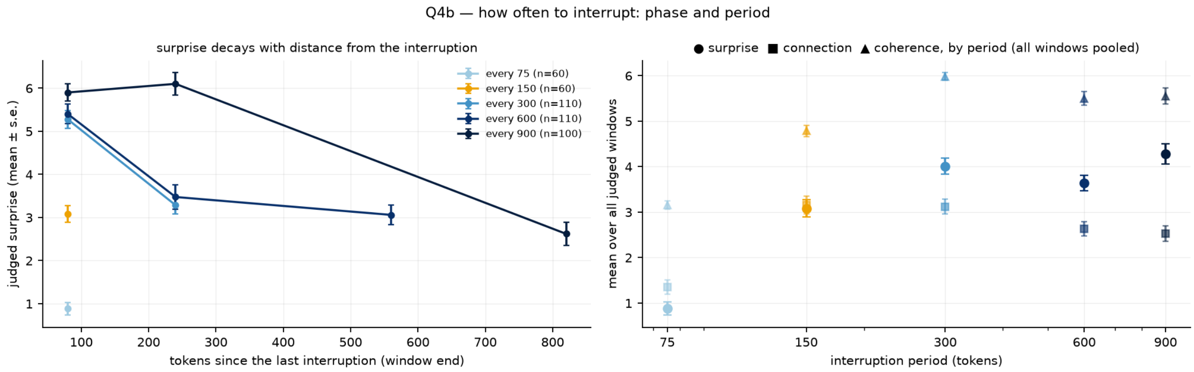

Figure 9: How often to interrupt. Left: judged surprise against tokens since the last interruption, per period. Right: stream mean by period. Surprise peaks in the 150 tokens after a break and decays; a break every 75 tokens develops nothing.

(0.88); connection falls as segments lengthen (Figure 9). The loop's rhythm is: let a thread develop for a few hundred tokens, then break it away.

6.4 Three generator families

The ladder replicates on Qwen3-8B-Base and OLMo-2-13B (Table 2, Figures 10–11). On the 8B: $0.68 \to 1.47 \to 2.73$ (connection $0.87 \to 3.23$, $\delta = +0.80$), scaffold 2.73 / 1.53. On OLMo, whose bare stream wanders across web genres rather than locking into literal orbits and is therefore less dead (1.35), the same ordering holds on surprise and connection ($2.37 \to 3.11$; connection $1.51 \to 3.17$, $\delta = +0.58$), with one new nuance: the plain interruption *costs* coherence there (-0.77 [-1.58, +0.03]; the injected narrative seed lands on 4,000 tokens of footers) and the scaffold's selective forgetting is what preserves it (4.48 vs 3.28), the mechanism the calibration probes had suggested. Whether forgetting is worth its connection cost depends on how deep the generator's well is.

7 Inside the network

Judges and sentence embeddings see the text; the model's own residual stream sees the computation. We re-run every finished stream through its generator (one forward pass with a KV cache, exact token positions) and capture the residual at 13 layers (every 4th of 48 for Qwen3-30B-A3B; every 3rd for the 36-layer Qwen3-8B and the 40-layer OLMo-2), mean-pooled over

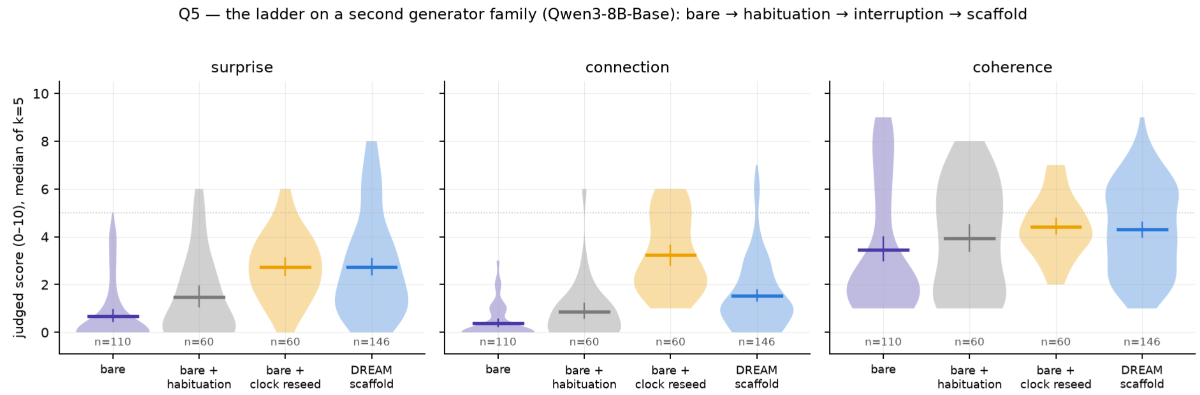

Figure 10: The ladder on Qwen3-8B-Base.

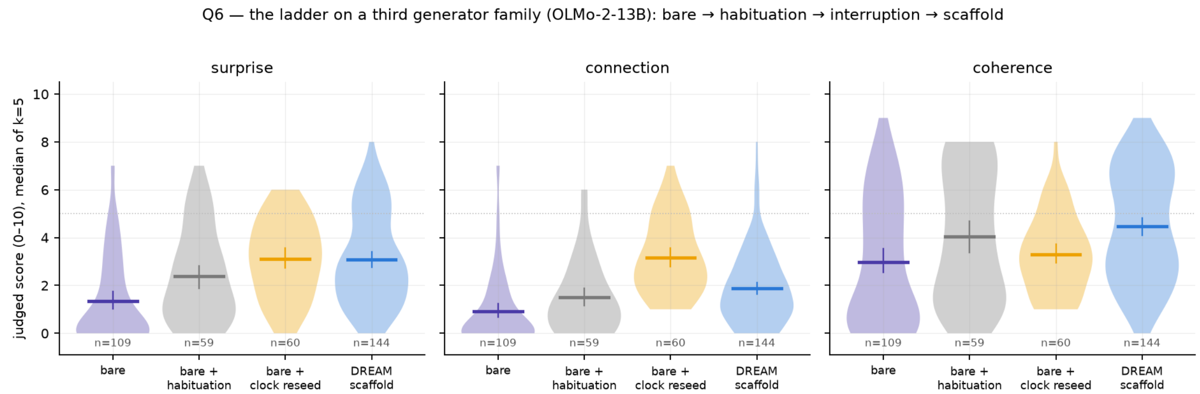

Figure 11: The ladder on OLMo-2-13B: same ordering; the plain interruption costs coherence and forgetting preserves it.

64-token windows (stride 32) and mean-centered per layer before cosine geometry, plus a logit lens at each captured layer (the final norm and unembedding applied to the intermediate state). Three questions.

N1: where does the stream freeze? Per-layer trajectory geometry of the window vectors (mean step between consecutive windows, explored radius) by condition, and the logit-lens *commitment layer*, the captured layer from which the top-1 token no longer changes. Bare generation's mean step is a third to a quarter of every interrupted condition's at every layer; its explored radius is 0.13 at the input layers against 0.42–0.53, and the gap narrows with depth but never closes (0.24 vs 0.33–0.34 at layer 47, CIs disjoint; Figure 13). The three interrupted conditions are geometrically alike inside the network; what separates them is what the judge reads. On the 30B, bare generation's top-1 stabilizes *later* (commitment index 11.12 [11.04, 11.25] of 12 vs 10.88–10.96) while its final distribution is far more confident (final entropy 0.34 vs 0.42–1.08), the signature of copying, a late-layer computation that ends certain; the ordering does not replicate on the 8B, so we report it as an observation. The 8B and OLMo replicate the geometry (bare radius 0.14 and 0.36 at layer 0 against 0.46–0.61).

N2: which layer's movement predicts judged surprise? For every judged window, its novelty at layer l (cosine distance between the window's mean state and the mean state of everything before it) and its local step (vs the previous 160 tokens), correlated with judged surprise (Figure 14). Pooled over conditions, novelty against the past correlates with surprise at the input layers ($\rho = +0.47$ at layer 0) and not at the top (-0.04), but the pooled number

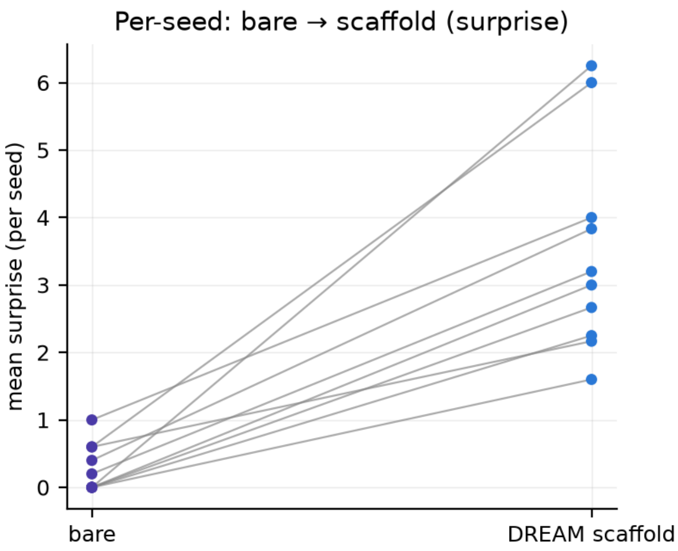

Figure 12: Per-seed robustness: mean surprise of bare vs scaffold windows, one line per seed.

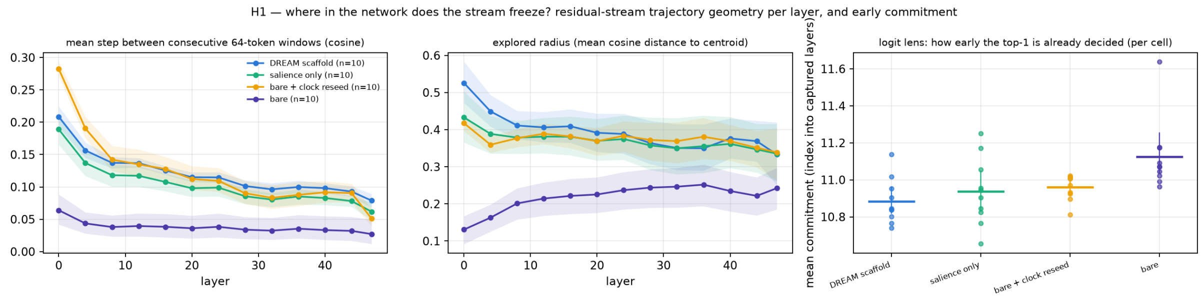

Figure 13: N1: residual-stream trajectory geometry per layer and condition (Qwen3-30B-A3B, ablation battery): mean step, explored radius, and logit-lens commitment. Bare generation freezes at every layer, most at the surface.

is carried by the bare/interrupted contrast. Within interrupted conditions the sign flips with depth: local step at layer 0 correlates positively with surprise (salience-only +0.46, scaffold +0.23) and at layer 47 negatively (−0.34, −0.14); novelty against the whole past is negative at depth (salience-only −0.53 at layer 47). Across the 668 judged windows of battery 2, layer-0 movement predicts surprise within every condition (ρ +0.3 to +0.6), and higher final entropy predicts it everywhere (ρ +0.3 to +0.6): confident is unsurprising. What the judge calls surprise is surface departure with deep continuity: new at the surface, continuous underneath.

N3: how deep does an interruption reach? Cosine distance between the 64 tokens before an injection and the 64 after it (the injected text skipped), per layer, minus the same at random positions; and the change in similarity to the premise's own state (Figure 15). Across a scaffold reseed *with forgetting*, the state after the injection differs from the state before by +0.09 (layer 4) growing to +0.21 (layer 40) beyond control, and its similarity to the premise's state rises by +0.09 to +0.20 with depth: forgetting is a deep re-encounter with the beginning, in the network's own representation. Across a clock reseed *over preserved context* the same measures are +0.01–0.03 and ≈0 at every layer, and this is the interruption the judge rewards most. Judged surprise and connection are not deep representational shifts; they are surface departures over an intact deep context, which is also why this arm keeps the most connection. The dissociation replicates on the 8B (+0.03 to +0.10 vs +0.01–0.03) and, larger, on OLMo (+0.20 to +0.38 with a return

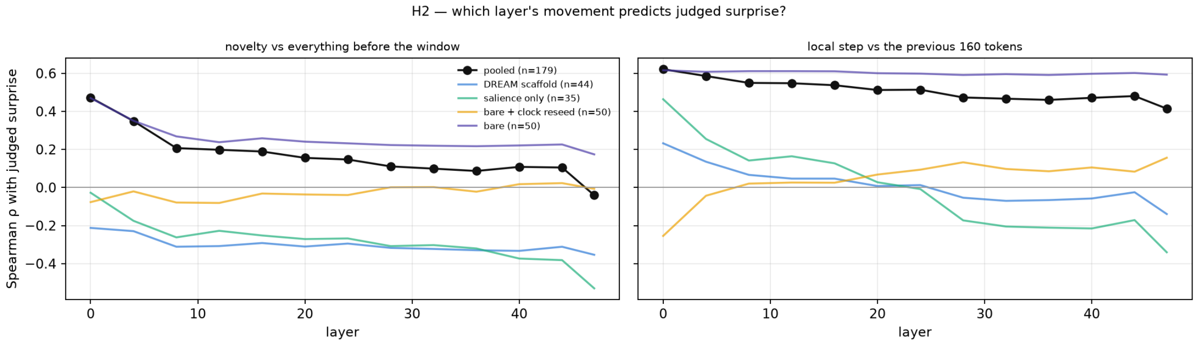

Figure 14: N2: Spearman correlation between the judged window's movement (novelty vs everything before it, left; local step, right) at each layer and judged surprise, pooled and within condition.

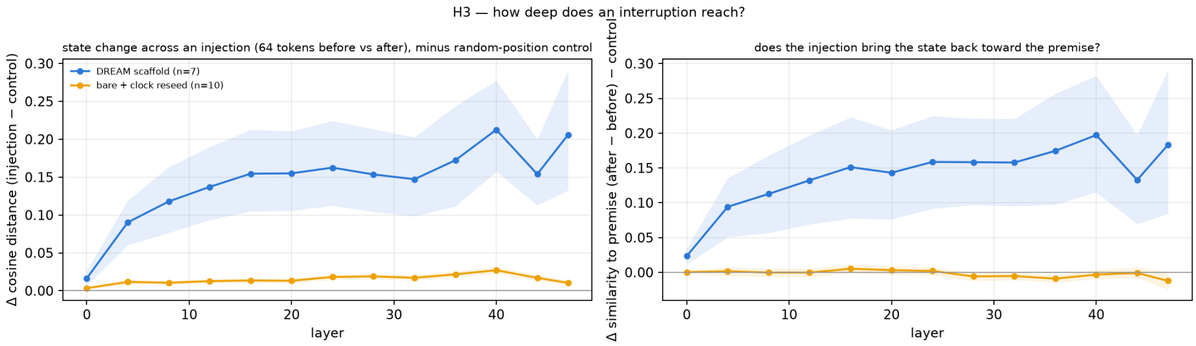

Figure 15: N3: how deep an interruption reaches (Qwen3-30B-A3B). Left: state change across an injection per layer, minus random-position control. Right: change in similarity to the premise's own state. The forgetting reseed moves the deep state and returns it toward the premise; the clock reseed over preserved context barely touches it, and is what the judge rewards.

of +0.13 to +0.18, vs +0.02–0.06 with no return). In battery 2, the depth of the state change scales with the length of the interrupted segment (period 75: ≈0; 150: +0.01–0.02; 300: +0.02–0.04; 600: +0.04–0.075) and not with what is injected (the internal counterpart of the frequency result), and the salience-timed stitch is the one 150-scale injection that reliably moves the deep state and pulls it toward the premise, while being the arm whose stream the judge rates lowest: deep movement and judged quality dissociate here too. Along the stream, similarity to the premise state is U-shaped in depth (≈0.9 at layer 0, ≈0.5 mid-network, rising at the top) and ordered at the top layer scaffold 0.85 > salience-only 0.79 > clock reseed 0.67 > bare 0.55: the sentence-embedding "return to the premise" of the scaffold is a top-layer phenomenon.

8 The instrument, measured

Every idea-level claim in this paper rests on an LLM judge, so we measured the judge before using it. On the same 89 windows judged $k = 5$ times by two models, Claude Opus 5 gives continuous scores with an intra-window spread of 0.71 (resolution at about ±0.7); Claude Sonnet 5 has spread 0.27, but by scoring zero on the connection dimension almost everywhere; a ruler that only reads "0" is consistent (Figure 16). The three-dimension surprise/connection/coherence rubric has spreads of 0.45–0.70 under Opus. The consequences shape the paper: LLM judges detect effects of one to two points (scaffold vs bare, salience vs clock, habituation vs interruption) and cannot adjudicate half-point effects (push vs plain), which is why the sampler's null inside the loop is reported as "undetectable at this resolution", not "absent"; and a binary threshold on a noisy judge is a coin (the same window scored 5.24 one night and 4.48 the next), so every

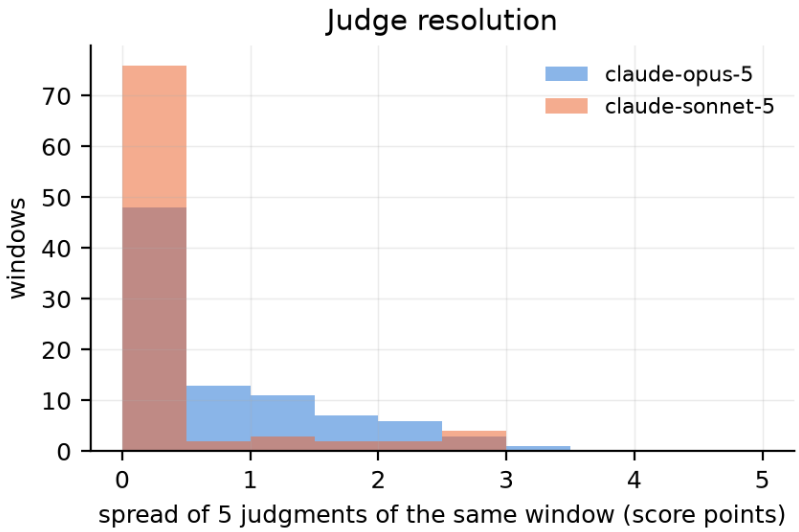

Figure 16: Spread of five judgments of the same window (delta rubric, 89 windows). Sonnet 5 concentrates at zero spread by scoring zero; Opus 5 gives continuous scores at ± 0.7 noise.

result here uses continuous medians of k samples.

A second judge family. To check that the loop results are not the taste of one model family, a stratified sample of 140 already-judged windows (20 per condition across bare, habituation, clock reseed, scaffold, re-encounter stitch, period 300 and salience-timed re-encounter) was re-judged by Kimi K2.6 (Moonshot, via OpenRouter) with the same rubric and $k = 5$. Agreement with Opus 5 on the median of five: surprise Spearman $\rho = +0.85$, connection $+0.77$, coherence $+0.71$ ($n = 140$, $p < 10^{-22}$). Kimi is more generous (means about one point higher) and noisier (intra-window spread 2.0–2.6), but every condition ordering replicates: bare 0.55 $\rightarrow$ habituation 3.15 $\rightarrow$ clock reseed 4.30 $\approx$ scaffold 4.10 $\rightarrow$ period 300 5.65 on surprise, and the neutral reseed and the re-encounter stitch highest on connection under both judges.

Human rating. A blind rating of 63 windows (nine per condition, stratified by judged surprise, shuffled, no labels, the earlier text shown) by independent raters recruited on a freelance platform without knowledge of the hypotheses or conditions: so far three of four (a native English reader with professional editing experience, a native/bilingual English speaker, and an advanced non-native reader with graduate training in language; the fourth, a bilingual editor, is pending). Each was given a one-page guide with the judge's rubric and instructed not to use AI tools or web searches; they rated machine-generated text only, no personal data was collected, and they were paid a fixed fee of R$187–300 (roughly US$35–55) for about 90 minutes of work. All three returned complete ratings using the whole of both scales, with distinct personal calibrations (mean surprise 3.4, 4.3 and 6.1). Each rater alone agrees with Opus 5 on surprise (Spearman $\rho = +0.40$, $+0.50$, $+0.47$; all $p \leq 10^{-3}$) and on coherence ($+0.60$, $+0.47$, $+0.23$); the mean of the three agrees more strongly than any one (surprise $\rho = +0.58$, $p = 7 \times 10^{-7}$; coherence $+0.52$, $p = 10^{-5}$), about as strongly as the raters agree with each other (pairwise Spearman on surprise 0.71, 0.38, 0.25; Krippendorff's interval $\alpha = 0.30$; on coherence 0.69, 0.60, 0.42; $\alpha = 0.38$; the low α reflects the different scale calibrations that rank correlation ignores). The human consensus reproduces the condition ordering that carries the paper: period-300 interruption 6.5 $\approx$ neutral clock reseed 6.4 $>$ re-encounter stitch 5.3 $>$ salience-timed 5.0 $>$ scaffold 3.6 $\approx$ habituation-only 3.6 $>$ bare 1.5 (surprise, means over nine windows); interruption vs bare $[+3.1, +6.5]$, habituation vs bare $[+0.1, +3.9]$, and the plain interruption above the full scaffold by a wider margin than under Opus. On coherence the human readers rank habituation without interruption highest (7.7) and the interrupted arms 5.7–6.3: to a human reader, the subject change inside a window costs some fluency that the LLM judge does not penalize. [update with the fourth rater]

9 Discussion

The scaffold and the beam. We started from an architecture modeled on the mind (three networks, incubation, re-encounter), and measurement returned something simpler and stronger than the architecture. The DREAM scaffold works, but when it is taken apart almost all of its effect lives in two operations that were not the noble ones: not eating one's literal past, and being made, every few hundred tokens, to start a new sentence somewhere else while remembering everything. We call this minimal recipe the *interrupted loop*. The elaborate parts either add nothing (kicks, escalation), or cost something (forgetting costs connection), or work opposite to their design (a literal return to the premise closes the loop; salience is a good reader of the stream and a bad metronome for it). Only because the whole scaffold was built and then dismantled do we know which beam carries the weight, and which pieces are conditional: forgetting earns its keep on a generator whose well is web boilerplate, and not on one that drifts in prose.

Surface novelty and idea novelty are different quantities. The anti-probable sampler doubles n-gram novelty against the training corpus and does nothing to judged surprise or connection; the interruption does nothing special to n-grams and lifts judged surprise from 0.3 to 3–6. Inside the network the two live at different depths: what the judge calls surprise is lexical departure over an intact deep context, not a deep representational shift, and the interruption that works barely moves the deep state. This is also why the neat expectation "novelty = distance from what came before" fails: within interrupted streams, deep departure from the stream's own past is *negatively* related to judged surprise. The reader rewards the window that is new on the surface and continuous underneath, which is what a good turn in a story is.

Rhythm. The frequency result is, we think, the most usable one: a stream cut every 75 tokens develops nothing; the same break that scores 3.1 after a 150-token thread scores 5.3–5.9 after a 300–900-token thread; and the yield decays over the next few hundred tokens. Surprise needs a developed background to break, and interruption is a resource with a rate. The rhythm we found, develop for a few hundred tokens and then break away, is the incubation structure in miniature: leave, and be made to come back on your own.

What this is not. It is not a claim that these streams are good literature; most windows are judged well below the midpoint, and the texts are what an unsupervised base model produces. It is not a claim about scale; the generators are 8–30B parameters at 8 bits. It is a claim about *where* the novelty a base model can produce with no task comes from, made with controls, three generator families and two judge families, and it points at the loop's rhythm rather than at the two places where effort is usually spent.

10 Limitations

Ten seeds, all English narrative premises. Three generator families for the loop's core ladder, one (Qwen3-30B-A3B) for the content, timing and frequency battery. LLM judges: Opus 5, calibrated and agreeing with a second family (Kimi K2.6, $\rho = 0.71$–0.85), but not yet with humans; the blind rating by independent readers is in progress. The judged sample of each cell is six windows per review kind, evenly spread; salience-timed arms are judged on selected moments and on a uniform grid, and both are reported, because the first is a selection. The residual-stream analysis is correlational, uses mean-pooled 64-token windows at 13 layers, and does not touch attention or causal interventions. The sampler's null inside the loop is a null at the instrument's resolution (≈1 point). Whether the rhythm (≈300 tokens) and the "lead away, then let it come back on its own" rule hold for other genres and for tasks with a verifier is the next question, and the fusion the program points to: an interrupted loop over a *problem*, with a verifier in place of the judge.

11 Conclusion

Where does novelty come from when a language model writes with no task? Not from the sampler: an anti-probable decoder buys surface novelty that never rises to the level of ideas. Not from the prompt: improbable inputs are noise. It comes from the loop, and from a simple part of it. A base model feeding on its own output freezes; keep it from eating its literal past and interrupt it, every few hundred tokens, with a sentence that leads away, over everything it remembers, and it comes alive: it develops, is broken, and finds its own way back, and that is what a reader (human, or one of two model families) calls surprise with connection. Inside the network the working interruption is a surface event over an intact deep state; the deep re-encounter is forgetting, and it is rewarded less. Creativity, in this system, is in interrupting the loop.

Acknowledgements and tooling. The experiments, analyses and text of this paper were produced by the author working with Claude (Anthropic) as a programming and writing assistant inside the Claude Code environment; every design decision, result and claim was reviewed by the author, and the dated laboratory notebook, code and per-run data are public in the repository. Judges were run on Amazon Bedrock and OpenRouter; generators ran locally on Apple silicon via MLX.

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A Full tables

The tables below are generated from the run data by the analysis scripts (`scripts/analysis.py`, `analysis_b2.py`, `hidden_analysis.py`) and typeset by `scripts/appendix_tex.py`; the complete set, including every per-layer table of the residual-stream analysis, ships with the code as `docs/APPENDIX_*.md`. Unless stated, the unit is the judged window (median of $k = 5$ Opus 5 judgments), windows before 100 generated tokens are excluded, and intervals are 95% bootstrap CIs.

dimension	condition	n	mean ± sd	scaffold − cond CI95	Cliff's δ	Mann-Whitney p
surprise	DREAM scaffold (λ=0)	47	3.19 ± 1.92	–	–	–
surprise	salience only	38	2.63 ± 1.75	[-0.22, +1.33]	+0.16	0.195
surprise	bare + clock reseed	60	3.08 ± 1.48	[-0.54, +0.78]	+0.00	0.967
surprise	bare generation	50	0.28 ± 0.72	[+2.34, +3.50]	+0.88	4.58e-15
connection	DREAM scaffold (λ=0)	47	1.64 ± 1.56	–	–	–
connection	salience only	38	1.39 ± 1.42	[-0.40, +0.87]	+0.09	0.446
connection	bare + clock reseed	60	3.15 ± 1.53	[-2.09, -0.91]	-0.58	1.52e-07
connection	bare generation	50	0.20 ± 0.57	[+0.98, +1.93]	+0.65	7.08e-10
coherence	DREAM scaffold (λ=0)	47	3.68 ± 1.69	–	–	–
coherence	salience only	38	3.71 ± 1.60	[-0.72, +0.67]	-0.03	0.819
coherence	bare + clock reseed	60	4.78 ± 0.95	[-1.65, -0.56]	-0.42	0.00014
coherence	bare generation	50	2.06 ± 2.01	[+0.87, +2.35]	+0.57	5.34e-07

Table 3: Ablation battery (Qwen3-30B-A3B, 10 seeds, Opus 5 k=5): scaffold vs each condition, unit = window.

dimension	windows	mean spread	p90 spread
surprise	242	0.61	1.00
connection	242	0.60	1.00
coherence	242	0.68	1.00

Table 4: Instrument calibration: intra-window spread of k=5 judgments per dimension (Opus 5).

arm	n	novel 4-grams %	novel 6-grams %	Δ4 vs baseline CI95	Cliff's δ	max copied span (words)
min-p baseline	15	21.5 ± 10.1	70.9 ± 11.9	–	–	10
λ=1	15	36.0 ± 12.4	87.6 ± 7.4	[+0.069, +0.230]	+0.68	10
λ=2	15	45.5 ± 10.8	89.7 ± 5.9	[+0.163, +0.314]	+0.88	13

Table 5: Phase 1: objective novelty against the OLMo-2 training corpus (3 prompts × 5 seeds).

arm	runs	train-plateau escapes	mean champion test excess	best
plain	5	2/5	0.0602	0.0600
antiprob	5	3/5	0.0604	0.0601

Table 6: Verified search on Qwen3-30B-A3B (bin packing): the anti-probable operator does not separate from plain sampling.

condition	windows	kind	surprise	connection	coherence	good (S≥5&H≥5)
bare (no habituation, no interruption)	50	jump	0.28 ± 0.72	0.20 ± 0.57	2.06 ± 2.01	0/50
bare + habituation (no interruption)	60	clock	1.70 ± 1.71	1.08 ± 1.10	4.08 ± 2.19	4/60
bare + habituation + clock reseed 150	60	clock	3.08 ± 1.48	3.15 ± 1.53	4.78 ± 0.95	12/60
DREAM scaffold (λ=0)	47	crystallize/jump/recurrence	3.19 ± 1.92	1.64 ± 1.56	3.68 ± 1.69	8/47

Table 7: Battery 2, Q1: habituation and interruption (Qwen3-30B-A3B).

condition	dim	Δ [CI]	δ	p	paired-by-seed Δ [CI] (n seeds)
bare (no habituation, no interruption)	surprise	-1.42 [-1.89, -0.95]	-0.58	1.4e-08	-1.42 [-1.72, -1.12] (10)
bare (no habituation, no interruption)	connection	-0.88 [-1.20, -0.56]	-0.52	1.1e-07	-0.88 [-1.22, -0.54] (10)
bare (no habituation, no interruption)	coherence	-2.02 [-2.79, -1.23]	-0.55	3.4e-07	-2.02 [-3.37, -0.68] (10)
bare + habituation + clock reseed 150	surprise	+1.38 [+0.80, +1.93]	+0.50	1.3e-06	+1.38 [+0.92, +1.92] (10)
bare + habituation + clock reseed 150	connection	+2.07 [+1.60, +2.55]	+0.75	5.2e-13	+2.07 [+1.63, +2.57] (10)
bare + habituation + clock reseed 150	coherence	+0.70 [+0.10, +1.30]	+0.21	4.3e-02	+0.70 [-0.27, +1.72] (10)
DREAM scaffold (λ=0)	surprise	+1.49 [+0.80, +2.19]	+0.46	3.4e-05	+1.80 [+0.97, +2.75] (10)
DREAM scaffold (λ=0)	connection	+0.55 [+0.06, +1.10]	+0.20	6.2e-02	+0.61 [+0.22, +1.01] (10)
DREAM scaffold (λ=0)	coherence	-0.40 [-1.14, +0.34]	-0.10	3.8e-01	-0.26 [-1.03, +0.56] (10)

Table 8: Battery 2, Q1: habituation and interruption (Qwen3-30B-A3B); differences vs the reference arm (Δ mean with 95% CI over windows; Cliff's δ; Mann–Whitney p; paired-by-seed Δ [CI]).

condition	windows	kind	surprise	connection	coherence	good (S≥5&H≥5)
neutral subject change (clock 150)	60	clock	3.08 ± 1.48	3.15 ± 1.53	4.78 ± 0.95	12/60
re-encounter stitch (clock 150)	50	cut	2.68 ± 1.46	3.64 ± 1.57	4.72 ± 1.33	5/50
the premise itself (clock 150)	50	cut	1.14 ± 1.73	1.04 ± 1.57	3.18 ± 1.90	3/50
the stream's own past (clock 150)	50	cut	1.42 ± 1.43	1.52 ± 1.35	3.06 ± 1.42	2/50

Table 9: Battery 2, Q2: content of the interruption at period 150.

condition	dim	Δ [CI]	δ	p	paired-by-seed Δ [CI] (n seeds)
re-encounter stitch (clock 150)	surprise	-0.40 [-0.95, +0.14]	-0.13	2.4e-01	-0.40 [-1.29, +0.46] (10)
re-encounter stitch (clock 150)	connection	+0.49 [-0.09, +1.07]	+0.20	7.1e-02	+0.49 [-0.56, +1.56] (10)
re-encounter stitch (clock 150)	coherence	-0.06 [-0.50, +0.38]	-0.00	9.8e-01	-0.06 [-0.75, +0.57] (10)
the premise itself (clock 150)	surprise	-1.94 [-2.53, -1.31]	-0.68	6.1e-10	-1.94 [-2.65, -1.30] (10)
the premise itself (clock 150)	connection	-2.11 [-2.68, -1.51]	-0.72	4.1e-11	-2.11 [-2.83, -1.46] (10)
the premise itself (clock 150)	coherence	-1.60 [-2.15, -1.01]	-0.53	1.2e-06	-1.60 [-2.36, -0.70] (10)
the stream's own past (clock 150)	surprise	-1.66 [-2.21, -1.11]	-0.61	2.1e-08	-1.66 [-2.26, -1.07] (10)
the stream's own past (clock 150)	connection	-1.63 [-2.17, -1.08]	-0.62	9.0e-09	-1.63 [-2.37, -0.91] (10)
the stream's own past (clock 150)	coherence	-1.72 [-2.18, -1.24]	-0.70	1.1e-10	-1.72 [-2.30, -1.11] (10)

Table 10: Battery 2, Q2: content of the interruption at period 150; differences vs the reference arm (Δ mean with 95% CI over windows; Cliff's δ; Mann–Whitney p; paired-by-seed Δ [CI]).

condition	windows	kind	surprise	connection	coherence	good (S≥5&H≥5)
re-encounter on the clock (150)	50	cut	2.68 ± 1.46	3.64 ± 1.57	4.72 ± 1.33	5/50
re-encounter on the clock (900; matched frequency)	40	cut	2.73 ± 1.95	2.30 ± 1.82	4.67 ± 1.59	6/40
neutral change on the clock (900; matched frequency)	40	cut	2.48 ± 1.87	2.02 ± 1.68	4.95 ± 2.16	5/40
re-encounter on salience events (event windows)	80	crystallize/cut/jump/recurrence	3.92 ± 1.42	2.99 ± 1.73	3.75 ± 1.18	8/80
re-encounter on salience events (uniform windows)	60	clock	2.18 ± 2.04	1.68 ± 1.65	3.93 ± 1.88	4/60
re-encounter on the clock (900), uniform windows	60	clock	4.70 ± 1.69	3.48 ± 1.52	5.50 ± 1.41	31/60
neutral change on the clock (900), uniform windows	60	clock	5.48 ± 1.64	2.87 ± 1.56	5.95 ± 1.32	45/60
salience only, no injection (event windows)	38	crystallize/jump/recurrence	2.63 ± 1.75	1.39 ± 1.42	3.71 ± 1.60	2/38

Table 11: Battery 2, Q3: timing of the re-encounter, clock vs salience.

condition	dim	Δ [CI]	δ	p	paired-by-seed Δ [CI] (n seeds)
re-encounter on the clock (900; matched frequency)	surprise	+0.04 [-0.68, +0.78]	-0.03	8.1e-01	+0.04 [-0.53, +0.66] (10)
re-encounter on the clock (900; matched frequency)	connection	-1.34 [-2.03, -0.62]	-0.45	2.2e-04	-1.34 [-2.17, -0.53] (10)
re-encounter on the clock (900; matched frequency)	coherence	-0.04 [-0.65, +0.58]	-0.02	8.9e-01	-0.05 [-0.87, +0.73] (10)
neutral change on the clock (900; matched frequency)	surprise	-0.21 [-0.90, +0.51]	-0.10	4.4e-01	-0.21 [-0.84, +0.36] (10)
neutral change on the clock (900; matched frequency)	connection	-1.62 [-2.28, -0.92]	-0.51	2.5e-05	-1.61 [-2.17, -1.13] (10)
neutral change on the clock (900; matched frequency)	coherence	+0.23 [-0.56, +0.99]	+0.11	3.5e-01	+0.23 [-0.45, +1.07] (10)
re-encounter on salience events (event windows)	surprise	+1.24 [+0.73, +1.75]	+0.44	2.0e-05	+1.10 [+0.34, +1.85] (10)
re-encounter on salience events (event windows)	connection	-0.65 [-1.23, -0.06]	-0.24	2.2e-02	-0.92 [-1.55, -0.29] (10)
re-encounter on salience events (event windows)	coherence	-0.97 [-1.41, -0.52]	-0.39	1.5e-04	-0.86 [-1.72, -0.06] (10)
re-encounter on salience events (uniform windows)	surprise	-0.50 [-1.14, +0.15]	-0.21	5.5e-02	-0.50 [-1.11, -0.01] (10)
re-encounter on salience events (uniform windows)	connection	-1.96 [-2.54, -1.35]	-0.61	2.7e-08	-1.96 [-2.62, -1.37] (10)
re-encounter on salience events (uniform windows)	coherence	-0.79 [-1.38, -0.19]	-0.25	2.1e-02	-0.79 [-1.31, -0.29] (10)
re-encounter on the clock (900), uniform windows	surprise	+2.02 [+1.43, +2.60]	+0.64	5.9e-09	+2.02 [+1.62, +2.44] (10)
re-encounter on the clock (900), uniform windows	connection	-0.16 [-0.74, +0.43]	-0.03	7.7e-01	-0.16 [-0.68, +0.37] (10)
re-encounter on the clock (900), uniform windows	coherence	+0.78 [+0.29, +1.30]	+0.31	4.4e-03	+0.78 [+0.40, +1.15] (10)
neutral change on the clock (900), uniform windows	surprise	+2.80 [+2.23, +3.37]	+0.77	1.9e-12	+2.80 [+2.04, +3.59] (10)
neutral change on the clock (900), uniform windows	connection	-0.77 [-1.37, -0.18]	-0.27	1.3e-02	-0.77 [-1.78, +0.23] (10)
neutral change on the clock (900), uniform windows	coherence	+1.23 [+0.74, +1.73]	+0.48	7.2e-06	+1.23 [+0.56, +1.84] (10)
salience only, no injection (event windows)	surprise	-0.05 [-0.73, +0.64]	-0.05	7.0e-01	+0.18 [-0.80, +1.31] (10)
salience only, no injection (event windows)	connection	-2.25 [-2.87, -1.62]	-0.73	3.8e-09	-2.11 [-3.00, -1.16] (10)
salience only, no injection (event windows)	coherence	-1.01 [-1.64, -0.37]	-0.35	4.5e-03	-0.87 [-1.92, +0.12] (10)

Table 12: Battery 2, Q3: timing of the re-encounter, clock vs salience; differences vs the reference arm (Δ mean with 95% CI over windows; Cliff's δ; Mann–Whitney p; paired-by-seed Δ [CI]).

condition	windows	kind	surprise	connection	coherence	good (S≥5&H≥5)
every 75 (before injection)	50	cut	0.86 ± 1.04	1.28 ± 1.04	3.08 ± 0.63	1/50
every 150 (before injection)	60	clock	3.08 ± 1.48	3.15 ± 1.53	4.78 ± 0.95	12/60
every 300 (before injection)	50	cut	2.90 ± 1.50	2.70 ± 1.63	5.88 ± 0.82	7/50
every 600 (before injection)	50	cut	3.06 ± 1.58	2.56 ± 1.66	5.32 ± 1.62	6/50
every 900 (before injection)	40	cut	2.48 ± 1.87	2.02 ± 1.68	4.95 ± 2.16	5/40
every 300 (mid-segment)	60	clock	4.93 ± 1.53	3.47 ± 1.68	6.07 ± 0.98	34/60
every 600 (mid-segment)	60	clock	4.12 ± 1.83	2.70 ± 1.63	5.65 ± 1.54	23/60
every 900 (mid-segment)	60	clock	5.48 ± 1.64	2.87 ± 1.56	5.95 ± 1.32	45/60

Table 13: Battery 2, Q4: frequency of interruption (neutral change).

condition	dim	Δ [CI]	δ	p	paired-by-seed Δ [CI] (n seeds)
every 75 (before injection)	surprise	-2.22 [-2.69, -1.75]	-0.83	1.7e-14	-2.22 [-2.91, -1.73] (10)
every 75 (before injection)	connection	-1.87 [-2.35, -1.38]	-0.76	1.1e-12	-1.87 [-2.49, -1.32] (10)
every 75 (before injection)	coherence	-1.70 [-2.00, -1.41]	-0.86	1.5e-15	-1.70 [-2.00, -1.39] (10)
every 300 (before injection)	surprise	-0.18 [-0.74, +0.37]	-0.06	5.7e-01	-0.18 [-0.81, +0.39] (10)
every 300 (before injection)	connection	-0.45 [-1.04, +0.12]	-0.18	1.1e-01	-0.45 [-1.31, +0.45] (10)
every 300 (before injection)	coherence	+1.10 [+0.78, +1.42]	+0.59	2.5e-08	+1.10 [+0.69, +1.57] (10)
every 600 (before injection)	surprise	-0.02 [-0.60, +0.54]	-0.00	9.8e-01	-0.02 [-0.74, +0.60] (10)
every 600 (before injection)	connection	-0.59 [-1.19, +0.01]	-0.21	4.9e-02	-0.59 [-1.28, +0.03] (10)
every 600 (before injection)	coherence	+0.54 [+0.03, +1.03]	+0.26	1.8e-02	+0.54 [-0.09, +1.13] (10)
every 900 (before injection)	surprise	-0.61 [-1.28, +0.08]	-0.21	6.7e-02	-0.61 [-1.41, +0.18] (10)
every 900 (before injection)	connection	-1.12 [-1.78, -0.47]	-0.38	1.1e-03	-1.12 [-2.11, -0.26] (10)
every 900 (before injection)	coherence	+0.17 [-0.56, +0.88]	+0.12	3.1e-01	+0.17 [-0.79, +1.12] (10)
every 300 (mid-segment)	surprise	+1.85 [+1.30, +2.38]	+0.60	9.0e-09	+1.85 [+1.23, +2.33] (10)
every 300 (mid-segment)	connection	+0.32 [-0.27, +0.88]	+0.13	2.0e-01	+0.32 [-0.30, +0.77] (10)
every 300 (mid-segment)	coherence	+1.28 [+0.93, +1.62]	+0.63	1.1e-09	+1.28 [+0.88, +1.68] (10)
every 600 (mid-segment)	surprise	+1.03 [+0.45, +1.62]	+0.33	1.3e-03	+1.03 [+0.58, +1.47] (10)
every 600 (mid-segment)	connection	-0.45 [-1.03, +0.10]	-0.17	9.1e-02	-0.45 [-1.00, +0.17] (10)
every 600 (mid-segment)	coherence	+0.87 [+0.40, +1.32]	+0.42	4.6e-05	+0.87 [+0.32, +1.43] (10)
every 900 (mid-segment)	surprise	+2.40 [+1.83, +2.93]	+0.70	1.5e-11	+2.40 [+1.98, +2.78] (10)
every 900 (mid-segment)	connection	-0.28 [-0.83, +0.27]	-0.09	3.8e-01	-0.28 [-0.75, +0.10] (10)
every 900 (mid-segment)	coherence	+1.17 [+0.75, +1.57]	+0.56	7.2e-08	+1.17 [+0.78, +1.57] (10)

Table 14: Battery 2, Q4: frequency of interruption (neutral change); differences vs the reference arm (Δ mean with 95% CI over windows; Cliff's δ; Mann–Whitney p; paired-by-seed Δ [CI]).

every	windows	pooled surprise	conn	coh	early (≤160 tokens since injection) S / C / H	late (>160) S / C / H	phase-weighted stream mean S / C / H
75	60	0.88	1.35	3.15	0.88 / 1.35 / 3.15 (n=60)	, /: /: (n=0)	0.88 / 1.35 / 3.15
150	60	3.08	3.15	4.78	3.08 / 3.15 / 4.78 (n=60)	, /: /: (n=0)	3.08 / 3.15 / 4.78
300	110	4.01	3.12	5.98	5.28 / 3.35 / 5.95 (n=40)	3.29 / 2.99 / 6.00 (n=70)	4.35 / 3.18 / 5.97
600	110	3.64	2.64	5.50	5.40 / 3.20 / 5.65 (n=20)	3.24 / 2.51 / 5.47 (n=90)	3.82 / 2.69 / 5.52
900	100	4.28	2.53	5.55	5.90 / 2.80 / 6.10 (n=40)	3.20 / 2.35 / 5.18 (n=60)	3.68 / 2.43 / 5.35

Table 15: Battery 2, Q4b: phase within the segment and phase-weighted stream means.

condition	windows	kind	surprise	connection	coherence	good (S≥5&H≥5)
8B: bare	110	clock/jump	0.68 ± 1.26	0.38 ± 0.73	3.45 ± 2.64	1/110
8B: bare + habituation	60	clock	1.47 ± 1.69	0.87 ± 1.23	3.93 ± 2.26	4/60
8B: bare + habituation + clock reseed 150	60	clock/cut	2.73 ± 1.41	3.23 ± 1.60	4.42 ± 1.28	1/60
8B: DREAM scaffold (λ=0)	146	clock/crystallize/cut/jump	2.73 ± 2.13	1.53 ± 1.47	4.30 ± 1.88	15/146

Table 16: Second generator family (Qwen3-8B-Base).

condition	dim	Δ [CI]	δ	p	paired-by-seed Δ [CI] (n seeds)
8B: bare	surprise	-0.78 [-1.28, -0.30]	-0.30	3.0e-04	-0.78 [-1.13, -0.45] (10)
8B: bare	connection	-0.48 [-0.84, -0.16]	-0.27	7.1e-04	-0.48 [-0.84, -0.14] (10)
8B: bare	coherence	-0.49 [-1.23, +0.28]	-0.13	1.5e-01	-0.49 [-2.43, +1.58] (10)
8B: bare + habituation + clock reseed 150	surprise	+1.26 [+0.69, +1.80]	+0.47	6.3e-06	+1.26 [+0.52, +2.14] (10)
8B: bare + habituation + clock reseed 150	connection	+2.37 [+1.87, +2.87]	+0.80	9.4e-15	+2.37 [+1.68, +3.12] (10)
8B: bare + habituation + clock reseed 150	coherence	+0.48 [-0.18, +1.13]	+0.13	2.1e-01	+0.48 [-0.53, +1.43] (10)
8B: DREAM scaffold (λ=0)	surprise	+1.27 [+0.70, +1.81]	+0.38	1.8e-05	+1.17 [+0.62, +1.68] (10)
8B: DREAM scaffold (λ=0)	connection	+0.66 [+0.25, +1.04]	+0.31	2.7e-04	+0.67 [+0.27, +1.07] (10)
8B: DREAM scaffold (λ=0)	coherence	+0.36 [-0.30, +1.01]	+0.10	2.4e-01	+0.52 [-0.44, +1.27] (10)

Table 17: Second generator family (Qwen3-8B-Base); differences vs the reference arm (Δ mean with 95% CI over windows; Cliff's δ; Mann–Whitney p; paired-by-seed Δ [CI]).

condition	windows	kind	surprise	connection	coherence	good (S≥5&H≥5)
OLMo: bare	109	clock/jump	1.35 ± 1.80	0.91 ± 1.53	2.97 ± 2.65	6/109
OLMo: bare + habituation	59	clock	2.37 ± 1.91	1.51 ± 1.50	4.05 ± 2.68	7/59
OLMo: bare + habituation + clock reseed 150	60	clock/cut	3.11 ± 1.63	3.17 ± 1.54	3.28 ± 1.63	6/60
OLMo: DREAM scaffold (λ=0)	144	clock/crystallize/cut/jump	3.08 ± 1.94	1.89 ± 1.54	4.48 ± 2.32	22/144

Table 18: Third generator family (OLMo-2-13B).

condition	dim	Δ [CI]	δ	p	paired-by-seed Δ [CI] (n seeds)
OLMo: bare	surprise	-1.02 [-1.62, -0.44]	-0.36	7.8e-05	-1.02 [-2.08, -0.10] (10)
OLMo: bare	connection	-0.60 [-1.07, -0.13]	-0.30	4.8e-04	-0.60 [-1.52, +0.31] (10)
OLMo: bare	coherence	-1.08 [-1.92, -0.26]	-0.25	6.1e-03	-1.08 [-2.34, -0.01] (10)
OLMo: bare + habituation + clock reseed 150	surprise	+0.74 [+0.09, +1.36]	+0.26	1.5e-02	+0.72 [-0.30, +1.53] (10)
OLMo: bare + habituation + clock reseed 150	connection	+1.66 [+1.10, +2.20]	+0.58	3.0e-08	+1.65 [+0.77, +2.30] (10)
OLMo: bare + habituation + clock reseed 150	coherence	-0.77 [-1.58, +0.03]	-0.12	2.4e-01	-0.78 [-1.77, +0.25] (10)
OLMo: DREAM scaffold (λ=0)	surprise	+0.71 [+0.13, +1.29]	+0.22	1.4e-02	+0.81 [+0.18, +1.47] (10)
OLMo: DREAM scaffold (λ=0)	connection	+0.38 [-0.09, +0.84]	+0.17	5.6e-02	+0.49 [-0.02, +1.02] (10)
OLMo: DREAM scaffold (λ=0)	coherence	+0.43 [-0.34, +1.20]	+0.11	2.3e-01	+0.69 [-0.33, +1.63] (10)

Table 19: Third generator family (OLMo-2-13B); differences vs the reference arm (Δ mean with 95% CI over windows; Cliff's δ; Mann–Whitney p; paired-by-seed Δ [CI]).

B Reproducibility

Everything is in the repository (117 tests): sampler core, MLX adapter, salience monitor, reverie engine, judges, novelty client, experiment runners with resumable overnight execution and thermal logging; every run's per-step telemetry, texts, token streams and judgments; and the dated decision log. Judge cost of the entire program: about US$120.